

Math 122: Exam 1 Practice 1 Solutions

Problem 1 (Section 5.4)

Find the exact arc length of the curve $y = 3x^{3/2} - 1$ over the interval from $x = 0$ to $x = 1$.

Solution

The arc length formula for a function $y = f(x)$ on $[a, b]$ is:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

1. Find the derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = 3 \cdot \frac{3}{2} x^{1/2} = \frac{9}{2} x^{1/2}$$

2. Square the derivative:

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{9}{2} x^{1/2}\right)^2 = \frac{81}{4} x$$

3. Set up the integral:

$$L = \int_0^1 \sqrt{1 + \frac{81}{4} x} dx$$

4. Evaluate using substitution: Let $u = 1 + \frac{81}{4}x$, which gives $du = \frac{81}{4} dx \implies dx = \frac{4}{81} du$.

- When $x = 0$: $u = 1$
- When $x = 1$: $u = 1 + \frac{81}{4} = \frac{85}{4}$

$$\begin{aligned}
L &= \frac{4}{81} \int_1^{85/4} u^{1/2} du \\
&= \frac{4}{81} \left[\frac{2}{3} u^{3/2} \right]_1^{85/4} \\
&= \frac{8}{243} \left[\left(\frac{85}{4} \right)^{3/2} - 1 \right] \\
&= \frac{8}{243} \left(\frac{85\sqrt{85}}{8} - 1 \right) \\
&= \frac{85\sqrt{85} - 8}{243} \approx 3.1920
\end{aligned}$$

Problem 2 (Section 6.5)

Find the limit:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$$

Solution

Direct substitution produces the indeterminate form:

$$\frac{e^0 - 1}{\sin 0} = \frac{0}{0}$$

Apply L'Hôpital's Rule by differentiating the numerator and denominator:

$$\lim_{x \rightarrow 0} \frac{\frac{d}{dx}(e^x - 1)}{\frac{d}{dx}(\sin x)} = \lim_{x \rightarrow 0} \frac{e^x}{\cos x}$$

Evaluate the limit:

$$\frac{e^0}{\cos 0} = \frac{1}{1} = 1$$

Problem 3 (Section 6.5)

Find the limit:

$$\lim_{x \rightarrow +\infty} x e^{-x}$$

Solution

Rewrite the expression in fractional form:

$$\lim_{x \rightarrow +\infty} \frac{x}{e^x}$$

Direct evaluation yields the indeterminate form $\frac{\infty}{\infty}$. Applying L'Hôpital's Rule:

$$\lim_{x \rightarrow +\infty} \frac{\frac{d}{dx}(x)}{\frac{d}{dx}(e^x)} = \lim_{x \rightarrow +\infty} \frac{1}{e^x} = 0$$

Problem 4 (Section 6.5)

Find the limit:

$$\lim_{x \rightarrow 0} (e^x + x)^{1/x}$$

Solution

This is an indeterminate form of type 1^∞ . Let:

$$y = (e^x + x)^{1/x}$$

Take the natural logarithm of both sides:

$$\ln y = \frac{1}{x} \ln(e^x + x) = \frac{\ln(e^x + x)}{x}$$

Evaluate the limit of $\ln y$ as $x \rightarrow 0$ ($\frac{0}{0}$ indeterminate form):

$$\lim_{x \rightarrow 0} \frac{\ln(e^x + x)}{x}$$

Apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{\frac{e^x+1}{e^x+x}}{1} = \frac{\frac{e^0+1}{e^0+0}}{1} = 2$$

Since $\lim_{x \rightarrow 0} \ln y = 2$, exponentiate to obtain the final limit:

$$\lim_{x \rightarrow 0} (e^x + x)^{1/x} = e^2$$

Problem 5 (Section 6.7)

Find $\frac{dy}{dx}$ for:

$$y = \sin^{-1} \left(\frac{1}{x} \right)$$

Solution

Using the chain rule and the derivative formula $\frac{d}{du} [\sin^{-1} u] = \frac{1}{\sqrt{1-u^2}}$:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(1/x)^2}} \cdot \frac{d}{dx} \left(\frac{1}{x} \right)$$

Since $\frac{d}{dx} (x^{-1}) = -\frac{1}{x^2}$:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{1}{x^2}}} \cdot \left(-\frac{1}{x^2} \right)$$

Simplify the square root:

$$\sqrt{1-\frac{1}{x^2}} = \sqrt{\frac{x^2-1}{x^2}} = \frac{\sqrt{x^2-1}}{|x|}$$

Multiply through:

$$\frac{dy}{dx} = -\frac{1}{x^2 \cdot \frac{\sqrt{x^2-1}}{|x|}} = -\frac{1}{|x|\sqrt{x^2-1}}$$

Problem 6 (Section 6.7)

Find $\frac{dy}{dx}$ for:

$$y = e^x \sec^{-1} x$$

Solution

Apply the Product Rule:

$$\frac{dy}{dx} = \frac{d}{dx}(e^x) \cdot \sec^{-1} x + e^x \cdot \frac{d}{dx}(\sec^{-1} x)$$

Using $\frac{d}{dx}(e^x) = e^x$ and $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$:

$$\frac{dy}{dx} = e^x \sec^{-1} x + \frac{e^x}{|x|\sqrt{x^2-1}} = e^x \left(\sec^{-1} x + \frac{1}{|x|\sqrt{x^2-1}} \right)$$

Problem 7 (Section 6.7)

Evaluate the integral:

$$\int \frac{dx}{\sqrt{1-4x^2}}$$

Solution

Write $4x^2$ as $(2x)^2$:

$$\int \frac{dx}{\sqrt{1-(2x)^2}}$$

Let $u = 2x$, so $du = 2 dx \implies dx = \frac{1}{2} du$:

$$\begin{aligned} \int \frac{\frac{1}{2} du}{\sqrt{1-u^2}} &= \frac{1}{2} \int \frac{du}{\sqrt{1-u^2}} \\ &= \frac{1}{2} \sin^{-1}(u) + C \\ &= \frac{1}{2} \sin^{-1}(2x) + C \end{aligned}$$

Problem 8 (Section 6.7)

Evaluate the integral:

$$\int \frac{\sec^2 x dx}{\sqrt{1-\tan^2 x}}$$

Solution

Let $u = \tan x$, then $du = \sec^2 x \, dx$:

$$\begin{aligned}\int \frac{du}{\sqrt{1-u^2}} &= \sin^{-1}(u) + C \\ &= \sin^{-1}(\tan x) + C\end{aligned}$$

Problem 9 (Section 6.7)

Evaluate the definite integral:

$$\int_1^{\sqrt{3}} \frac{\sqrt{\tan^{-1} x}}{1+x^2} dx$$

Solution

Let $u = \tan^{-1} x$, which means $du = \frac{1}{1+x^2} dx$.

Find the transformed bounds: - When $x = 1$: $u = \tan^{-1}(1) = \frac{\pi}{4}$ - When $x = \sqrt{3}$: $u = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

Substitute and integrate:

$$\begin{aligned}\int_{\pi/4}^{\pi/3} u^{1/2} du &= \left[\frac{2}{3} u^{3/2} \right]_{\pi/4}^{\pi/3} \\ &= \frac{2}{3} \left[\left(\frac{\pi}{3} \right)^{3/2} - \left(\frac{\pi}{4} \right)^{3/2} \right] \\ &= \frac{2\pi\sqrt{\pi}}{3} \left(\frac{\sqrt{3}}{9} - \frac{1}{8} \right)\end{aligned}$$

Problem 10 (Section 6.7)

Evaluate the integral:

$$\int \frac{\sin \theta}{\cos^2 \theta + 1} d\theta$$

Solution

Let $u = \cos \theta$, so $du = -\sin \theta d\theta \implies \sin \theta d\theta = -du$:

$$\begin{aligned}\int \frac{-du}{u^2 + 1} &= -\int \frac{du}{1 + u^2} \\ &= -\tan^{-1}(u) + C \\ &= -\tan^{-1}(\cos \theta) + C\end{aligned}$$